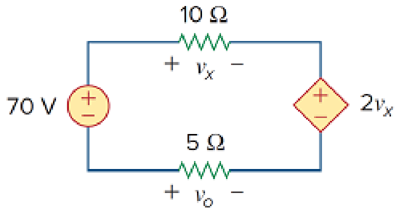
Chapter 2, Problem 6PP

Problem

Find *vx* and *vo* in the circuit of Fig. 2.24.

Figure 2.24:



Step-by-step solution

Step 1 of 4

The circuit is shown in Figure 1.

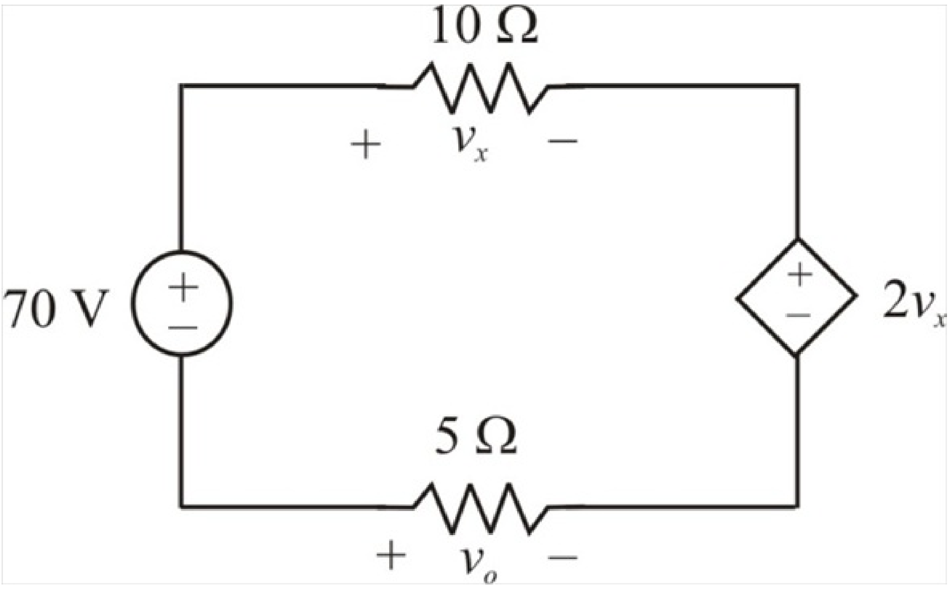


Figure 1

Step 2 of 4

Assume that the current i flows through the loop as shown in Figure 2.

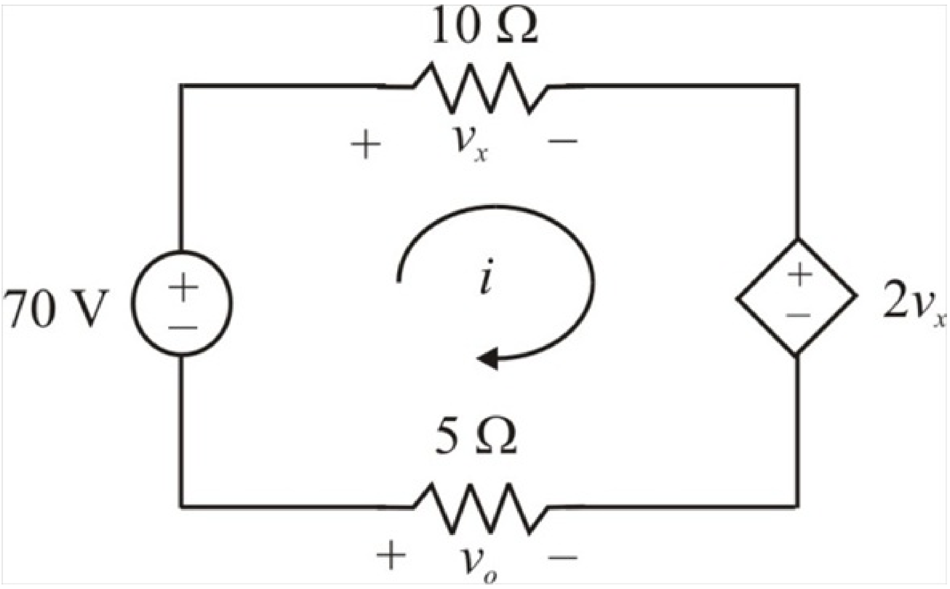


Figure 2

Step 3 of 4

To find v_x and v_o, we apply Ohm's law and Kirchhoff's voltage law.

From Ohm's law,

$v_x = 10i$ …… (1)

$v_o = -5i$ …… (2)

Applying Kirchhoff's voltage law around the loop gives,

$$-70 + v_x + 2v_x - v_o = 0$$
$$-70 + 10i + 20i - (-5i) = 0$$
$$35i = 70$$
$$i = 2\text{ A}$$

Substituting the value of i in equation (1), we get,

$$\begin{aligned} v_x &= 10i \\ &= (10)(2) \\ &= 20\text{ V} \end{aligned}$$

Therefore, the voltage v_x is $\boxed{20\text{ V}}$.

Step 4 of 4

Substituting the value of i in equation (2), we get,

$$\begin{aligned} v_o &= -5i \\ &= (-5)(2) \\ &= -10\text{ V} \end{aligned}$$

Therefore, the voltage v_o is $\boxed{-10\text{ V}}$.